Grade yourself: formula written? numbers substituted? arithmetic shown? answer boxed to 4 decimals? interpretation sentence where asked? Each part on the real exam is scored on those steps, not only the final number.

Question 1 (6 points) — Commuting and Housing

First add the margins: Drive 42, Bus 26, Walk 26, Bike 16 (row totals); on campus 49, off campus 61; overall 110.

(a) one category → its total ÷ grand total

$$P(\text{off campus}) = \frac{61}{110} = 0.5545$$

0.5545

(b) a row → add its cells, ÷ 110

$$P(\text{walk}) = \frac{22 + 4}{110} = \frac{26}{110} = 0.2364$$

0.2364

(c) OR → add, subtract the overlap (drive AND on campus = 8) once

$$P(\text{drive or on}) = \frac{42}{110} + \frac{49}{110} - \frac{8}{110} = \frac{83}{110} = 0.7545$$

0.7545

(d) AND → the one cell ÷ 110

$$P(\text{bike and off}) = \frac{7}{110} = 0.0636$$

0.0636

(e) GIVEN off campus → divide by the off-campus total (61), not 110

$$P(\text{bus} \mid \text{off}) = \frac{16}{61} = 0.2623$$

0.2623

Common errors: (c) forgetting the overlap gives 91/110 = 0.8273; (e) 16/110 = 0.1455 uses the wrong denominator.

Question 2 (2 points) — Valid distribution?

Rule 1: 0.15, 0.25, 0.30, 0.20, 0.12 are all between 0 and 1 ✓. Rule 2: add them.

$$0.15 + 0.25 + 0.30 + 0.20 + 0.12 = 1.02 \neq 1$$

NOT a valid probability distribution — the probabilities add to 1.02, not 1.

Write both checks; the question says "explain why."

Question 3 (8 points) — Cars per Household

(a) P(x) = frequency ÷ total; total = 12 + 28 + 32 + 8 = 80

$$P(0) = \frac{12}{80} = 0.1500 \qquad P(1) = \frac{28}{80} = 0.3500 \qquad P(2) = \frac{32}{80} = 0.4000 \qquad P(3) = \frac{8}{80} = 0.1000$$

Check: 0.1500 + 0.3500 + 0.4000 + 0.1000 = 1 ✓

(b) x across (0, 1, 2, 3), P(x) up; a vertical bar above each x with heights 0.15, 0.35, 0.40, 0.10; tallest at x = 2; label both axes.

(c)

$$\mu = \sum x\,P(x) = 0(0.15) + 1(0.35) + 2(0.40) + 3(0.10) = 0 + 0.35 + 0.80 + 0.30 = 1.45$$

μ = 1.45 cars

Interpretation: over many households in this neighborhood, the average number of cars per household is about 1.45.

(d) $\sigma = \sqrt{\Sigma(x-\mu)^2 \cdot P(x)}$ with $\mu = 1.45$

$$(0-1.45)^2(0.15) = 2.1025(0.15) = 0.3154 \qquad (1-1.45)^2(0.35) = 0.2025(0.35) = 0.0709$$
$$(2-1.45)^2(0.40) = 0.3025(0.40) = 0.1210 \qquad (3-1.45)^2(0.10) = 2.4025(0.10) = 0.2403$$
$$\sigma^2 = 0.3154 + 0.0709 + 0.1210 + 0.2403 = 0.7476 \qquad \sigma = \sqrt{0.7476} = 0.86$$

σ ≈ 0.86 cars

(Unrounded: σ² = 0.7475, σ = 0.8646; either rounds to 0.86.)

Question 4 (8 points) — Reading Habits (binomial, n = 8, p = 0.35, 1 − p = 0.65)

$$P(x) = {}_8C_x\,(0.35)^x(0.65)^{8-x}$$

(a) exactly 2

$${}_8C_2 = \frac{8\cdot 7}{2\cdot 1} = 28 \qquad P(2) = 28(0.35)^2(0.65)^6 = 28(0.1225)(0.0754) = 0.2587$$

0.2587

(b) at most 2 → x = 0, 1, 2 (the 2 is IN)

$$P(0) = (0.65)^8 = 0.0319 \qquad P(1) = 8(0.35)(0.65)^7 = 0.1373 \qquad P(2) = 0.2587$$
$$P(X \le 2) = 0.0319 + 0.1373 + 0.2587 = 0.4279 \approx 0.4278$$

0.4278 (0.4279 if you add the rounded values — both accepted)

(c) at least 1 → 1 − P(0)

$$P(X \ge 1) = 1 - (0.65)^8 = 1 - 0.0319 = 0.9681$$

0.9681

(d) n = 300

$$\mu = np = 300(0.35) = 105 \qquad \sigma = \sqrt{np(1-p)} = \sqrt{300(0.35)(0.65)} = \sqrt{68.25} = 8.26$$

μ = 105, σ ≈ 8.26

Exact values: P(2) = 0.2587, P(0) = 0.0319, P(1) = 0.1373, P(X ≤ 2) = 0.4278.

Question 5 (5 points) — Shuttle Wait Time (uniform, a = 0, b = 25)

(a)

$$\text{height} = \frac{1}{b-a} = \frac{1}{25-0} = \frac{1}{25} = 0.04$$

a flat line (rectangle) at height 1/25 from 0 to 25 minutes, and 0 elsewhere; area = 25 × 1/25 = 1

(b)

$$P(5 \le X \le 15) = \frac{15-5}{25-0} = \frac{10}{25} = 0.4000$$

0.4000

(c) "more than 18" → from 18 up to the top end 25

$$P(X > 18) = \frac{25 - 18}{25 - 0} = \frac{7}{25} = 0.2800$$

0.2800

Question 6 (6 points) — Normal Distribution and z-Scores

(a) right = 1 − left; hint P(Z < 0.85) = 0.8023

$$P(Z > 0.85) = 1 - 0.8023 = 0.1977$$

0.1977

(b) between = (left of the bigger z) − (left of the smaller z); −0.25 is bigger than −1.30

$$0.4013 - 0.0968 = 0.3045$$

0.3045

(c)

$$z = \frac{300 - 250}{40} = 1.25 \ \rightarrow \ \text{left} = 0.8944 \qquad P(X > 300) = 1 - 0.8944 = 0.1056$$

0.1056

Question 7 (8 points) — Heights of Adult Men (μ = 69, σ = 2.8)

(a) taller than 74 → right tail

$$z = \frac{74 - 69}{2.8} = 1.79 \ \rightarrow \ \text{left} = 0.9633 \qquad P(X > 74) = 1 - 0.9633 = 0.0367$$

0.0367

(b) shorter than 65 → left area (the hint is already the left area)

$$z = \frac{65 - 69}{2.8} = -1.43 \qquad P(X < 65) = 0.0764$$

0.0764

(c) between 66 and 72

$$z_1 = \frac{66 - 69}{2.8} = -1.07 \rightarrow 0.1423 \qquad z_2 = \frac{72 - 69}{2.8} = 1.07 \rightarrow 0.8577$$

$$P(66 < X < 72) = 0.8577 - 0.1423 = 0.7154$$

0.7154

(d) 10th percentile → area to the LEFT = 0.10 → hint z = −1.28 (negative: lower tail)

$$x = \mu + z\sigma = 69 + (-1.28)(2.8) = 69 - 3.584 = 65.416 \approx 65.4$$

≈ 65.4 inches

Sanity checks: (a) must be below 0.5 (74 is above average); (d) must be below 69.